Graphical Abstract

TEMPO: Regime-Aware Operator Learning with Locally Adapted POD Bases

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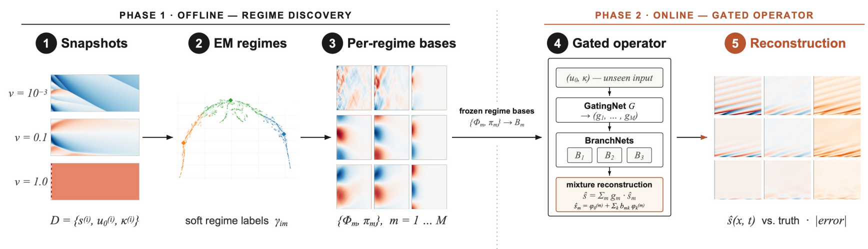

Highlights

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- TEMPO discovers latent solution regimes from unlabelled PDE snapshots via EM.
- Regime-local bases provably achieve no greater error than a global POD basis.
- TEMPO(POD) reduces error by 13-50
- Per-parameter specialists fail catastrophically out-of-distribution.
- Regime membership reflects initial-condition geometry, not the physical parameter.

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Abstract

Operator learning approximates the solution map of a parametric PDE from data, enabling fast prediction at new parameter values without repeated solves. The dominant approach fits a single global low-rank basis to all solution snapshots, an assumption that fails when the governing parameter drives qualitatively distinct solution regimes: smooth diffusive profiles versus near-discontinuous shocks, or solutions differing by orders of magnitude in amplitude. A single global basis must split its capacity across all regimes, faithfully representing none.

We propose TEMPO (Trajectory EM Mixture of POD Operators), a two-phase framework for regime-aware operator learning. In the offline phase, TEMPO models the trajectory ensemble as a Gaussian mixture and recovers M latent regimes via Expectation-Maximization: the E-step assigns each trajectory to regimes based on reconstruction quality, and the M-step fits a dedicated linear (POD) or nonlinear (NeuralPOD) basis to each. We prove that regime-local bases achieve reconstruction error no greater than the global rank-K optimum for any soft assignment matrix and any M, K. In the online phase, a GatingNet and M BranchNets are trained jointly, routing new inputs to the correct regime at inference.

We evaluate TEMPO on three benchmarks. On 1D Burgers' equation, TEMPO(POD) reduces error by 13–50% over the jointly-trained POD-DeepONet baseline across all viscosity values. On 2D Darcy flow, per-parameter specialists fail catastrophically out-of-distribution, while TEMPO(POD) reduces error by up to 5×. On 2D Navier-Stokes, TEMPO matches the global POD baseline. The discovered regime partition recovers structure driven by both initial conditions and the physical parameter from solution trajectories, without regime-label supervision.

Keywords: operator learning, parametric PDE, regime discovery, proper orthogonal decomposition, expectation-maximization, neural operator

1. Introduction

Reduced-order modeling of parametric PDEs constructs low-rank bases from solution snapshots to enable fast prediction at new parameter values without re-solving the governing equations [1]. The dominant approach, exemplified by POD-DeepONet [2], fits a single global basis to the full snapshot ensemble and trains a branch network to predict expansion coefficients for unseen inputs [3]. This works well when solutions lie on a smooth low-dimensional manifold, but fails when the governing parameter drives *qualitatively distinct* dynamics: viscosity separating laminar flow from near-discontinuous shocks, or source magnitude shifting solution amplitudes by orders of magnitude. A single global basis must distribute its capacity across all regimes, representing none faithfully.

Partitioning the parameter space and fitting local bases to each region is the classical remedy in reduced-order modeling [4, 5]. Daniel et al. [6] extend this to deep learning by clustering solution subspaces on the Grassmann manifold and training a dictionary selector at inference. These approaches reduce approximation error on multi-modal problems, but share a fundamental limitation: clustering is performed independently of basis quality, so the discovered partition need not align with the directions of greatest reconstruction benefit.

A second line of work enriches the basis while retaining a single global model. POD-PoU [7] blends several POD bases via a partition-of-unity function, reducing error on sharp-gradient problems. NeuralPOD [8] learns nonlinear mode functions by greedy residual minimisation, overcoming the linear subspace constraint of POD. Both improve representational capacity but treat the full ensemble jointly, without identifying physically distinct sub-populations.

Neural operators such as FNO [9] and CNO [10] learn solution operators as discretisation-invariant functional maps, achieving high accuracy without explicit basis decompositions. Their purpose is complementary to the present work: they do not produce inspectable basis representations or identify latent regime structure.

We propose **TEMPO** (*Trajectory EM Mixture of POD Operators*), which addresses the feedback limitation of local reduced-order modeling in a principled probabilistic framework. TEMPO models the trajectory ensemble as a Gaussian mixture and recovers regime structure via Expectation-Maximization (EM): the E-step softly assigns each trajectory to regimes based on reconstruction quality, while the M-step rebuilds a dedicated explicit basis per regime, so that basis quality sharpens regime assignments and regime assignments sharpen basis fitting iteratively. After convergence, a GatingNet and per-regime BranchNets are trained jointly, routing new inputs to the correct regime at inference.

Contributions:

- **TEMPO framework.** A two-phase mixture-of-operators for parametric PDEs: an offline EM that discovers M latent regimes from unlabelled snapshots and fits a dedicated orthogonal (POD) or nonlinear (NeuralPOD) basis to each; an online phase where a GatingNet and M BranchNets are trained with a KL penalty that anchors routing to the offline regime partition.

- **Adaptive basis update.** A distribution-shift criterion that monitors moment changes in regime composition across EM iterations and selects per-regime between full retraining, incremental mode addition/pruning, or skipping.
- **Theoretical guarantee.** Regime-local bases achieve reconstruction error no greater than the global rank-K optimum for any soft-assignment matrix and any M, K (Proposition 1).
- **Experiments.** On 1D Burgers' and 2D Darcy flow, TEMPO(POD) reduces error by 13–50% over the jointly-trained POD-DeepONet baseline, while per-parameter specialists fail badly out-of-distribution.

2. Problem Statement

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain and $\mathcal{T} \subset \mathbb{R}_{\geq 0}$ a time interval.

We are given a dataset of N solution trajectories:

$$\mathcal{D} = \left\{ \, (u_0^{(i)}, \, \kappa^{(i)}, \, s^{(i)}) \, \right\}_{i=1}^{N}, \qquad s^{(i)} = s\Big(\cdot, \, \cdot; \, u_0^{(i)}, \kappa^{(i)}\Big) \in L^2(\Omega \times \mathcal{T}), \quad (1)$$

where $u_0^{(i)} \in \mathcal{U} = L^2(\Omega)$ is the initial condition and $\kappa^{(i)} \in \mathcal{K}$ the physical parameter. Each trajectory $s^{(i)}$ is discretized on $N_y = N_t N_x$ quadrature points $\{y_j = (x_j, t_j), \, w_j\}_{j=1}^{N_y}$ covering the product grid $\Omega \times \mathcal{T}$, with weighted norm $\|f\|_w^2 = \sum_j w_j f(y_j)^2$.

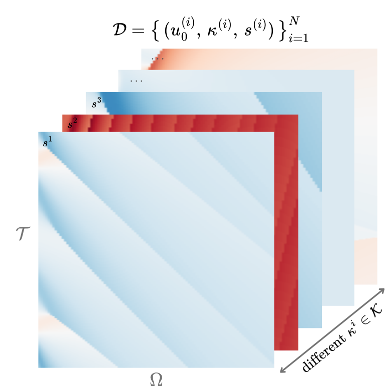

Figure 1: Training dataset $\mathcal{D}$: solution trajectories $s^{(i)}$ on $\Omega \times \mathcal{T}$ for different $\kappa^{(i)} \in \mathcal{K}$. Solutions change character fundamentally across parameter values.

Definition 1 (Solution operator)**.** *The* solution operator

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \to L^2(\Omega \times \mathcal{T})$$

maps each pair (u_0, κ) *to the full space-time solution:* $\mathcal{G}(u_0, \kappa) = s(\,\cdot\,, \,\cdot\,; \, u_0, \kappa)$.

The operator $\mathcal{G}$ is unknown and expensive to evaluate directly. The *operator learning problem* is to construct $\hat{\mathcal{G}} \approx \mathcal{G}$ from $\mathcal{D}$ alone, generalizing to unseen $(u_0, \kappa) \notin \mathcal{D}$ without solving the PDE.

Multi-regime assumption. When solutions change character fundamentally across $\mathcal{K}$, the optimal low-dimensional representation differs from regime to regime, and no single global basis captures the full solution diversity. We formalize this observation via a latent variable model over the trajectory ensemble.

Definition 2 (Generative model)**.** *Each trajectory arises from one of M latent regimes. The regime label $z_i \in \{1, \dots, M\}$ is unobserved and drawn from a categorical prior,*

$$z_i \sim \mathrm{Categorical}(\pi_1, \dots, \pi_M), \qquad \pi_m \geq 0, \quad \textstyle\sum_m \pi_m = 1. \tag{2}$$

Conditioned on $z_i = m$, the trajectory is distributed as

$$(s^{(i)} \mid z_i{=}m) \;\sim\; \mathcal{N}\big(\hat{s}_m^{(i)},\ \sigma^2 W^{-1}\big), \tag{3}$$

where $W = \mathrm{diag}(w_1, \dots, w_{N_y})$ is the quadrature weight matrix, $\sigma^2 > 0$ controls the softness of regime assignments, and $\hat{s}_m^{(i)}$ is the regime-m approximation to $s^{(i)}$.

The generative model leaves $\hat{s}_m^{(i)}$ unspecified. We require only that it admits a *low-rank trajectory decomposition*:

$$\hat{s}_m^{(i)}(y) \;=\; \phi_0^{(m)}(y) + \sum_{k=1}^{K_m} \lambda_k^{(m,i)}\, \phi_k^{(m)}(y), \qquad y = (x,t) \in \Omega \times \mathcal{T}, \tag{4}$$

where $\phi_k^{(m)} : \Omega \times \mathcal{T} \to \mathbb{R}$ are spatiotemporal mode functions and $\lambda_k^{(m,i)} \in \mathbb{R}$ a scalar amplitude for trajectory i on mode k. This form encompasses POD [2] and learned neural bases [8]; in this work we adopt the latter.

Definition 3 (Regime approximator)**.** *The parameters of regime m are $\Phi_m = \{\phi_k^{(m)}\}_{k=0}^{K_m}$ and scalar amplitudes $\Lambda_m = \{\lambda_k^{(m,i)}\}_{k=1,\, i=1}^{K_m,\, N}$, where $\phi_0^{(m)} : \Omega \times \mathcal{T} \to \mathbb{R}$ is the regime mean field, $\{\phi_k^{(m)}\}_{k\geq 1}$ spatiotemporal mode functions, $\lambda_k^{(m,i)} \in \mathbb{R}$ the scalar amplitude of trajectory i on mode k, and $K_m \leq K_{\max}$ the adaptive mode count.*

The learning problem decomposes into two phases: discovering the latent regime structure from $\mathcal{D}$ *offline*, then training a deployable operator for fast inference *online*.

Phase 1: Offline regime discovery. Since the regime labels z_i in (3) are unobserved, we recover the bases $\{\Phi_m\}$, mixture weights π, and noise level σ^2 by maximizing the observed-data log-likelihood:

$$(\Phi^*, \pi^*, \sigma^{2*}) \;=\; \operatorname*{arg\,max}_{\Phi,\, \pi,\, \sigma^2} \sum_{i=1}^{N} \log \sum_{m=1}^{M} \frac{\pi_m \, \det(W)^{1/2}}{(2\pi\sigma^2)^{N_y/2}} \exp\!\left(- \frac{\left\| s^{(i)} - \hat{s}_m^{(i)} \right\|_w^2}{2\sigma^2} \right). \tag{5}$$

The solution yields soft regime responsibilities $\gamma^*_{im} = p(z_i{=}m \mid s^{(i)}, \Phi^*, \pi^*, \sigma^{2*})$ that serve as supervision for Phase 2.

Phase 2: Online operator learning. With bases $\{\Phi^*_m\}$ and responsibilities $\{\gamma^*_{im}\}$ frozen, we train a gating network (*GatingNet*) and M branch networks (*BranchNets*) that map a new input (u_0, κ) to a weighted combination of local regime predictions. The precise architecture and loss are given in Section 3.4. The quality of the final operator is measured by the mean relative L^2 error:

$$\varepsilon \;=\; \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{\left\| s^{(i)} - \hat{s}(\cdot;\, u_0^{(i)}, \kappa^{(i)}) \right\|_w}{\left\| s^{(i)} \right\|_w}. \tag{6}$$

3. Method

All methods studied in this work share a common two-phase structure: an *offline* basis-fitting phase that constructs a low-rank approximation to the solution manifold from snapshots in $\mathcal{D}$, and an *online* operator-learning phase that trains a branch network to predict expansion coefficients from a new input (u_0, κ). The four methods differ along two independent axes:

- **Basis type.** A *linear* (POD) basis uses SVD of the (weighted) snapshot matrix; a *nonlinear* (NeuralPOD) basis learns spatiotemporal mode networks by sequential residual minimization.
- **Number of regimes.** A *single-regime* operator fits one global basis to all snapshots; a *multi-regime* operator (TEMPO) discovers M latent regimes via EM and assigns a dedicated basis to each.

This gives four combinations: POD-DeepONet [2] (linear, global), NeuralPOD-DeepONet (nonlinear, global, our extension of [8]), TEMPO(POD) (linear, M regimes), and TEMPO(NeuralPOD) (nonlinear, M regimes).

3.1. Basis Representations

3.1.1. POD Basis

Let $\phi_0 := \bar{s} = \frac{1}{N}\sum_i s^{(i)}$ be the mean field and $\tilde{S} \in \mathbb{R}^{N_y \times N}$ the matrix with columns $\tilde{s}^{(i)} = s^{(i)} - \phi_0$. The rank-$K$ truncated SVD $\tilde{S} \approx U_K \Sigma_K V_K^\top$ yields orthonormal modes $\{\phi_k\}_{k=1}^K$ (columns of U_K) and coefficients $\lambda_k^{(i)} = [\Sigma_K V_K^\top]_{ki}$; the approximation

$$\hat{s}^{(i)} = \phi_0 + \sum_{k=1}^{K} \lambda_k^{(i)}\, \phi_k \tag{7}$$

is optimal in the L^2 sense among all rank-K affine representations.

For TEMPO(POD), the M-step (17) is solved analytically: set $\phi_0^{(m)} := \bar{s}_m = \frac{1}{N_m}\sum_i \gamma_{im} s^{(i)}$, where $N_m = \sum_i \gamma_{im}$, and apply SVD to the weighted centered matrix with columns $\sqrt{\gamma_{im}}\,(s^{(i)} - \phi_0^{(m)})$. This is the closed-form linear analogue of Algorithm 1.

3.1.2. NeuralPOD Basis

To overcome the linearity constraint, we parametrize each mode function $\phi_k : \Omega\times\mathcal{T} \to \mathbb{R}$ as a Fourier feature network [11]; we call the resulting basis a *Fourier NeuralPOD* basis:

$$\phi_k(y) \;=\; f_{\theta_k}\big(\psi(y)\big), \qquad \psi(y) \;=\; \big[\sin(2\pi By),\, \cos(2\pi By)\big], \tag{8}$$

where $B \in \mathbb{R}^{F\times d}$ is a fixed random matrix drawn once at initialization and $f_{\theta_k} : \mathbb{R}^{2F} \to \mathbb{R}$ is a small MLP with Tanh activations. Plain MLPs exhibit spectral bias [12]; the Fourier encoding resolves this. For multi-scale coverage, the rows of B are partitioned evenly across L frequency scales $\sigma_1 < \cdots < \sigma_L$: the ℓ-th block is drawn from $\mathcal{N}(0,\, \sigma_\ell^2 I)$. The mean field ϕ_0 uses the same architecture.

The scalar amplitudes $\lambda_k^{(i)} \in \mathbb{R}$ are direct learnable parameters, one per training trajectory, optimized jointly with ϕ_k. After Phase 1 they are frozen and serve as regression targets for the BranchNet in Phase 2.

Modes are learned by *greedy residual minimization* [8]. Let $r^{(0,i)} = s^{(i)} - \phi_0$, where ϕ_0 is fitted to the snapshot mean $\bar{s}$ defined above. For each subsequent mode $k \geq 1$, given the residual

$$r^{(k-1,i)}(y) = s^{(i)}(y) - \phi_0(y) - \sum_{\ell=1}^{k-1} \lambda_\ell^{(i)}\, \phi_\ell(y), \tag{9}$$

we solve

$$\min_{\phi_k,\, \{\lambda_k^{(i)}\}} \sum_{i=1}^{N} \sum_{j} w_j \Big(\lambda_k^{(i)}\, \phi_k(y_j) - r_j^{(k-1,i)}\Big)^2 \tag{10}$$

(the global case with uniform weights; the γ-weighted generalization used in TEMPO is Algorithm 1). Mode addition stops when the residual falls below fraction τ of the initial variance,

$$\sum_{i=1}^{N} \|r^{(k,i)}\|_w^2 \;<\; \tau \cdot \sum_{i=1}^{N} \|s^{(i)} - \bar{s}\|_w^2, \tag{11}$$

or $k = K_{\max}$. Each mode captures the largest remaining variance component — the nonlinear analogue of Gram–Schmidt orthogonalization.

3.2. Single-Regime Operators

3.2.1. POD-DeepONet

POD-DeepONet [2] replaces the learned trunk network of DeepONet [3] with the fixed POD basis. After computing the global modes $\{\phi_k\}_{k=1}^{K}$ and training coefficients $\{\lambda^{(i)}\}$ offline, a branch network $\mathcal{B}_\theta : \mathbb{R}^{m_s+d_\kappa} \to \mathbb{R}^K$ is trained online by regression against these coefficients:

$$\mathcal{L}_{\text{branch}} = \frac{1}{N} \sum_{i=1}^{N} \big\|\mathcal{B}_\theta(u_0^{(i)},\, \kappa^{(i)}) - \lambda^{(i)}\big\|^2. \tag{12}$$

The prediction at any $(x,t) \in \Omega \times \mathcal{T}$ is

$$\hat{s}(x,t;\, u_0,\kappa) = \phi_0(x,t) + \sum_{k=1}^{K} \mathcal{B}_{\theta,k}(u_0,\kappa)\, \phi_k(x,t). \tag{13}$$

3.2.2. NeuralPOD-DeepONet

Mou et al. [8] introduce NeuralPOD and pair it with a DeepONet branch network to predict mode coefficients for unseen inputs. We adopt this combination, which we refer to as *NeuralPOD-DeepONet*, as a single-regime baseline and as the per-regime basis learner within TEMPO(NeuralPOD).

The procedure follows POD-DeepONet exactly with the Fourier NeuralPOD basis substituted for POD. Offline fitting yields coefficients $\boldsymbol{\lambda}^{(i)} \in \mathbb{R}^K$; online, a branch network minimizes (12) and prediction follows (13). At equal mode count K, the nonlinear basis spans a strictly richer function space than POD.

3.3. TEMPO: Offline Regime Discovery

A single global basis — whether linear or nonlinear — must compromise between qualitatively distinct regimes, as the optimal basis geometry differs fundamentally between them. TEMPO avoids this by constructing a dedicated basis for each of M latent regimes, discovered from unlabelled trajectories via EM on the generative model of Definition 2.

Initialization. A global POD is computed on all N trajectories; each trajectory is projected to a d_0-dimensional coefficient vector $\alpha^{(i)} \in \mathbb{R}^{d_0}$. Running a standard GMM on $\{\alpha^{(i)}\}$ yields initial responsibilities $\gamma_{im}^{(0)}$, weights $\pi_m^{(0)}$, and per-regime statistics $(\mu_m^{(0)}, \Sigma_m^{(0)})$. Each regime basis $\Phi_m^{(0)}$ is then fitted to the responsibility-weighted ensemble: for TEMPO(NeuralPOD) via NeuralBasis (Algorithm 1); for TEMPO(POD) via the weighted SVD of Section 3.1. The noise level σ^2 is calibrated from the initial reconstructions rather than treated as a free hyperparameter:

$$\sigma^2 \;\leftarrow\; \frac{1}{2N} \sum_{i=1}^{N} \sum_{m=1}^{M} \gamma_{im}^{(0)} \left\| s^{(i)} - \hat{s}_m^{(i)} \right\|_w^2, \tag{14}$$

σ^2 is held fixed throughout EM; improving bases reduce residuals, which progressively sharpens regime assignments.

E-step. At iteration q, the responsibility of trajectory i for regime m is updated by Bayes' rule on the Gaussian likelihood (3):

$$\gamma_{im}^{(q)} = \frac{\pi_m^{(q-1)} \exp\!\left(-\dfrac{\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2}{2\sigma^2}\right)}{\sum_{j=1}^{M} \pi_j^{(q-1)} \exp\!\left(-\dfrac{\|s^{(i)} - \hat{s}_j^{(i)}\|_w^2}{2\sigma^2}\right)}, \tag{15}$$

where $\hat{s}_m^{(i)}$ is evaluated via (4) using $\Phi_m^{(q-1)}$ and $\Lambda_m^{(q-1)}$. As the bases improve in the M-step, the reconstructions $\hat{s}_m^{(i)}$ sharpen and the E-step automatically refines regime assignments in response — distinguishing TEMPO from a GMM applied to a fixed feature space. When κ drives large amplitude variation (as with $\beta \in [0.01, 100]$ in the Darcy experiments), replacing the squared norm with the relative squared norm $\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2 / \|s^{(i)}\|_w^2$ throughout (in (15), (14), and (17)) keeps E- and M-steps consistent under a heteroscedastic likelihood with noise variance proportional to $\|s^{(i)}\|_w^2$. When κ is observed, the GMM initialization can operate in $\log \kappa$ space, and an optional κ-prior term in the E-step anchors assignments to the known parameter structure.

M-step. With responsibilities fixed, the M-step gives a closed-form update for the mixture weights,

$$\pi_m^{(q)} \leftarrow \frac{N_m^{(q)}}{N}, \qquad N_m^{(q)} = \sum_{i=1}^{N} \gamma_{im}^{(q)}, \tag{16}$$

and a weighted reconstruction problem for each basis:

$$\min_{\Phi_m,\, \Lambda_m} \sum_{i=1}^{N} \gamma_{im}^{(q)} \, \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2, \tag{17}$$

solved by weighted SVD for TEMPO(POD) or NeuralBasis (Algorithm 1) for TEMPO(NeuralPOD). Since gradient descent solves the NeuralPOD subproblem without guaranteeing global optimality, TEMPO is a Generalized EM algorithm [13], which guarantees that the observed-data log-likelihood is non-decreasing at every iteration.

Proposition 1 (Regime Approximation Advantage)**.** *For any $\{\gamma_{im}\} \geq 0$ with $\sum_m \gamma_{im} = 1$, any $M \geq 1$, any $K \geq 1$:*

$$\varepsilon_{\text{TEMPO}}^{K} \;\leq\; \varepsilon_{\text{global}}^{K},$$

where

$$\varepsilon_{\text{global}}^{K} = \min_{\substack{P \text{ rank-}K \\ \mu}} \sum_i \|(I - P)(s^{(i)} - \mu)\|_w^2,$$

$$\varepsilon_{\text{TEMPO}}^{K} = \sum_m \min_{\substack{P_m \text{ rank-}K \\ \mu_m}} \sum_i \gamma_{im} \|(I - P_m)(s^{(i)} - \mu_m)\|_w^2$$

are the global and regime-wise optimal rank-K reconstruction errors. Proof: Appendix A.1.

Remark 1. *Both $\varepsilon_{\text{TEMPO}}^{K}$ and $\varepsilon_{\text{global}}^{K}$ use exactly K basis functions per sample at inference; TEMPO stores M regime-specific bases of rank K offline, but routes each sample through exactly one.*

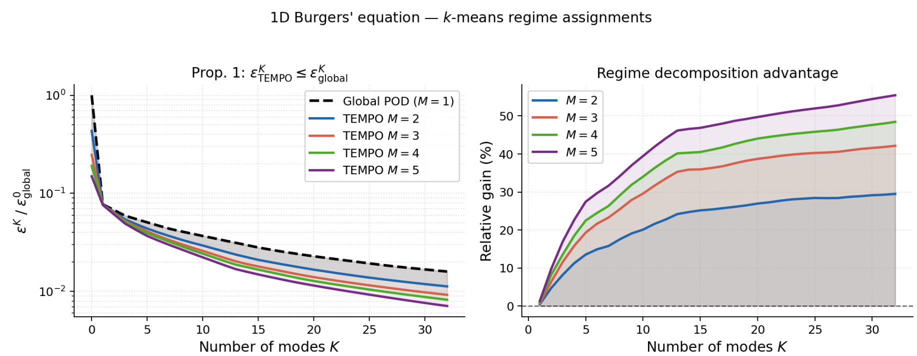

Figure 2: Empirical verification of Proposition 1 on 1D Burgers'. *Left*: normalized reconstruction error (log scale); TEMPO lies strictly below the global POD for all $K \geq 1$. *Right*: relative gain $(\varepsilon^K_{\text{global}} - \varepsilon^K_{\text{TEMPO}})/\varepsilon^K_{\text{global}}$; all values are non-negative, with gains of 20–39% at K=10.

Adaptive basis update. Retraining all M NeuralPOD bases at every EM iteration is computationally prohibitive. We introduce a distribution-shift criterion to determine whether each basis requires updating. Let $\alpha^{(i)}$ denote the global POD projection of $s^{(i)}$, computed once during initialization and fixed thereafter. The responsibility-weighted first and second moments of regime m are

$$\mu_m^{(q)} = \frac{1}{N_m^{(q)}} \sum_{i=1}^{N} \gamma_{im}^{(q)} \, \alpha^{(i)}, \tag{18}$$

$$\Sigma_m^{(q)} = \frac{1}{N_m^{(q)}} \sum_{i=1}^{N} \gamma_{im}^{(q)} \, (\alpha^{(i)} - \mu_m^{(q)})(\alpha^{(i)} - \mu_m^{(q)})^\top. \tag{19}$$

The distribution-shift criterion measures the relative change in these moments between consecutive iterations:

$$\Delta_m^{(q)} = \frac{\|\mu_m^{(q)} - \mu_m^{(q-1)}\|_2}{\|\mu_m^{(q-1)}\|_2 + \epsilon} + \frac{\|\Sigma_m^{(q)} - \Sigma_m^{(q-1)}\|_F}{\|\Sigma_m^{(q-1)}\|_F + \epsilon}. \tag{20}$$

where $\epsilon > 0$ is a small numerical stabilizer. $\Delta_m^{(q)}$ measures how much the composition of regime m has changed, not just the volume of reassignments. Two trajectories with very different $\alpha^{(i)}$ swapping regimes produce a large $\Delta_m^{(q)}$ and trigger a basis update; two similar snapshots doing so do not. Given thresholds $0 < \varepsilon_s < \varepsilon_l$, regime m is SKIPped if $\Delta_m^{(q)} < \varepsilon_s$, updated INCREMENTALly (add or prune one mode) if $\varepsilon_s \leq \Delta_m^{(q)} < \varepsilon_l$, and fully retrained from scratch (FULL RERUN) otherwise. In the INCREMENTAL case, we compute the residual of the current basis under the updated responsibilities $\gamma^{(q)}$, add a new mode if $\sum_i \gamma_{im}^{(q)} \|r_m^{(K_m,i)}\|_w^2 \geq \tau \cdot \sum_i \gamma_{im}^{(q)} \|s^{(i)} - \bar{s}_m^{(q)}\|_w^2$, and prune the last mode if its weighted contribution falls below ε_p. The FULL RERUN case uses a cold-start random initialization, as warm-starting may trap gradient descent in the geometry of the previous regime.

Convergence and model selection. The number of regimes M is selected by two criteria: BIC on the EM log-likelihood (penalizes complexity) and mean assignment entropy H. When they conflict, entropy is decisive: a large gain ΔH from M to $M{+}1$ signals a new regime. EM terminates when $\max_m \Delta_m^{(q)} < \varepsilon_c$. Selected values are $M{=}3$ (Burgers'), $M{=}4$ (Navier-Stokes), and $M{=}5$ (Darcy); details for Burgers' are in Table A.4 (Appendix A.3). The complete offline procedure is given in Algorithm 2. Figure 3 shows the resulting regime structure on 1D Burgers': three manifold branches emerge from unlabelled snapshots, one per discovered regime, with the partition reflecting joint variability from initial conditions and viscosity in the solution trajectories.

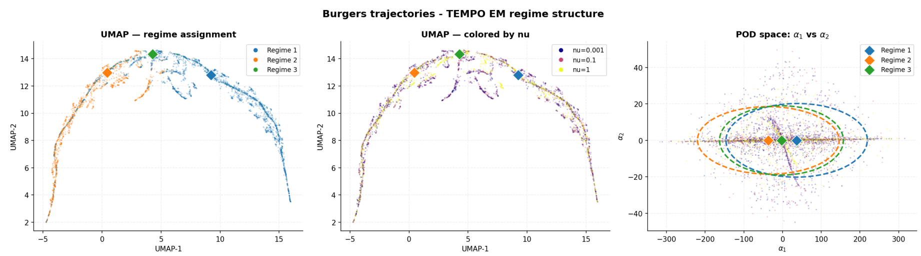

Figure 3: TEMPO(POD) regime structure on 1D Burgers' ($M{=}3$, 20 EM iterations). *Left:* UMAP colored by regime. *Center:* same embedding colored by ν. *Right:* GMM fit in POD coefficient space (α_1, α_2); stars mark regime centroids μ_m.

3.4. TEMPO: Online Gated Operator

After Algorithm 2 converges, all regime bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ are frozen. The online phase trains a GatingNet $G : \mathbb{R}^{m_s+d_\kappa} \to \Delta^{M-1}$ that maps (u_0, κ) to regime weights, and M BranchNets $\mathcal{B}_m : \mathbb{R}^{m_s+d_\kappa} \to \mathbb{R}^{K_m}$ that predict expansion coefficients for each regime. The TEMPO prediction at query point $y = (x,t) \in \Omega \times \mathcal{T}$ is

$$\hat{s}(y;\, u_0, \kappa) = \sum_{m=1}^{M} g_m(u_0, \kappa) \left[\phi_0^{(m)}(y) + \sum_{k=1}^{K_m} b_{m,k}(u_0, \kappa)\; \phi_k^{(m)}(y) \right], \qquad (21)$$

where $\mathbf{g} = G(u_0, \kappa)$ and $\mathbf{b}_m = \mathcal{B}_m(u_0, \kappa)$. The BranchNet outputs $\mathbf{b}_m$ predict the expansion coefficients $\boldsymbol{\lambda}_m^{(i)}$ for new inputs (u_0, κ) not seen in training. The prediction is mesh-free and evaluable at any $y \in \Omega \times \mathcal{T}$ without retraining.

Training objective. Writing $g_m^{(i)} = g_m(u_0^{(i)}, \kappa^{(i)})$, the online loss is

$$\begin{aligned}\mathcal{L}_{\text{online}} = &\underbrace{\frac{1}{N}\sum_{i=1}^{N} \|s^{(i)} - \hat{s}(\cdot;\, u_0^{(i)}, \kappa^{(i)})\|_w^2}_{\mathcal{L}_{\text{data}}} \\ &+\ \eta_{\text{KL}} \underbrace{\frac{1}{N}\sum_{i,m} g_m^{(i)} \log \frac{g_m^{(i)}}{\gamma_{im}^*}}_{\mathcal{L}_{\text{KL}}} \\ &-\ \eta_H \underbrace{\frac{1}{N}\sum_{i,m} g_m^{(i)} \log g_m^{(i)}}_{-\mathcal{L}_{\text{ent}}}.\end{aligned} \tag{22}$$

$\mathcal{L}_{\text{data}}$ drives reconstruction accuracy. $\mathcal{L}_{\text{KL}}$ anchors the online gating to the offline regime partition: it penalizes deviation of $\mathbf{g}^{(i)}$ from the EM responsibilities $\boldsymbol{\gamma}_i^*$, so the network cannot discard the discovered structure in pursuit of a simpler routing. $\mathcal{L}_{\text{ent}}$ prevents collapse to a single expert, ensuring all M regime bases contribute. The hyperparameters $\eta_{\text{KL}}, \eta_H \geq 0$ trade off these objectives.

4. Computational Experiments

We evaluate TEMPO on parametric PDE benchmarks, verifying three claims: (i) the offline EM phase recovers interpretable, physics-consistent regime structure from unlabelled snapshots; (ii) locally adapted bases achieve lower reconstruction error than a single global basis at equal mode count; (iii) a gated operator with per-regime bases outperforms a single global basis when the data contains structured regimes.

4.1. Experimental Setup

Datasets. We use three parametric PDE benchmarks; the first two are from PDEBench [14].

1D Burgers' equation.

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad x \in (-1, 1), \quad t \in (0, 2], \tag{23}$$

with periodic boundary conditions and random initial conditions. Viscosity $\nu \in \{0.001, 0.01, 0.1, 1.0\}$ spans four qualitatively distinct solution regimes: smooth diffusive profiles at ν=1.0 and near-discontinuous shock structures at ν=0.001. Each value provides N=9,500 trajectories on N_x=1,024 spatial points and N_t=201 time steps (38,000 total).

2D Darcy flow.

$$-\nabla \cdot \big(a(x)\,\nabla u(x)\big) = \beta, \qquad x \in (0,1)^2, \tag{24}$$

with zero Dirichlet boundary conditions. Here $a(x)$ is a spatially varying permeability field sampled from a random sum of Gaussians, and β is the constant source term taking five values: 0.01, 0.1, 1.0, 10.0, 100.0. Each value provides N=8,000 training and 1,000 test samples discretised on a 128×128 uniform grid (N_y=16,384 spatial points), 40,000 samples in total. The operator maps $a(x)$ to the pressure solution $u(x)$. As a steady-state problem, N_t=1 and there is no time dependence.

2D incompressible Navier-Stokes.

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \frac{1}{\mathrm{Re}}\,\Delta\omega + f, \qquad \mathbf{x} \in (0,1)^2,\ t \in (0,2], \tag{25}$$

with periodic boundary conditions, $\omega = \nabla \times \mathbf{u}$, divergence-free $\mathbf{u}$ recovered via the stream function, and Kolmogorov-type forcing $f(\mathbf{x}) = 0.1\big(\sin 2\pi(x_1{+}x_2) + \cos 2\pi(x_1{+}x_2)\big)$ [9]. Reynolds number Re $\in$ {100, 1000, 3600, 10000} spans four qualitatively distinct regimes from smooth laminar flow to developed turbulence. Each value provides N=5,000 trajectories on a 64×64 periodic grid (20,000 total; 4,000/1,000 train/test); see Appendix A.2. Operator: $\mathbf{u}_0 \mapsto (\mathbf{u}_t)_{t=1}^{20}$, Δt=0.1.

Methods. We compare DeepONet [3], POD-DeepONet [2], FNO [9], and CNO [10], all trained jointly on all parameter values via a single global model. We introduce NeuralPOD-DeepONet (learned Fourier NeuralPOD basis with branch network); TEMPO(POD) and TEMPO(NeuralPOD) (regime-aware bases discovered via EM, routed by gating network). Per-parameter specialists serve as performance ceilings. All methods implemented by the authors. Evaluation: mean relative L^2 error (6) per parameter value. Inference times measured on a single NVIDIA RTX A5000 GPU, per sample.

4.2. Main Results

1D Burgers' equation. Table 1 gives the cross-viscosity error matrix. Per-ν specialists are unviable across the full range: a ν=1.0 model degrades from 0.025 to 0.461 at ν=0.001. TEMPO(POD) addresses this by discovering regime structure jointly, outperforming the joint POD-DeepONet baseline by 13–50% across all ν. TEMPO(NeuralPOD) does not surpass the joint baseline. NeuralPOD coefficients are harder to predict than POD projections, as seen already in the single-regime case (0.229 vs. 0.048 at ν=0.1); iterative EM feedback compounds this.

2D Darcy flow. Table 2 gives the cross-β matrix. Specialist failure is catastrophic: Darcy solutions scale linearly with β, driving cross-parameter errors beyond 1000. TEMPO(POD) reduces error by 3–5× over the joint POD-DeepONet baseline (e.g., β=0.1: 0.775 vs. 2.557; β=100: 0.073 vs. 0.347). TEMPO(NeuralPOD) improves substantially over the global NeuralPOD-DeepONet baseline (β=0.1:

3.813 vs. 72.9) but does not match TEMPO(POD). FNO and CNO achieve lower absolute error by operating without explicit basis structure; the explicit regime partition and low-rank decomposition that TEMPO produces are not available from these methods.

2D Navier-Stokes. Table 3 gives the cross-Re error matrix. TEMPO(POD) matches the joint POD-DeepONet baseline within 5–17% but does not improve over it. Unlike Burgers' and Darcy, solutions at different Re share similar dominant POD directions: the EM assigns near-uniform regime weights, and local bases bring no reconstruction advantage over the global one. TEMPO(NeuralPOD) (κ-initialised, M=4) converges but does not close the gap to TEMPO(POD).

Inference speed. Routing each sample through a single regime basis keeps TEMPO(POD) fast. On Burgers', TEMPO(POD) requires 0.025 ms per sample versus 9.90 ms for FNO (400× faster) and 2.14 ms for CNO (86× faster); on Darcy, 0.006 ms versus 0.587 ms (FNO) and 2.24 ms (CNO). The overhead relative to POD-DeepONet is modest (0.025 vs. 0.004 ms on Burgers'), reflecting the additional gating network and M branch evaluations of which only one is active per sample.

4.3. Phase 1 Reconstruction Advantage

Figure 2 empirically verifies Proposition 1 on 1D Burgers' (N=1,500, $\nu \in \{0.001, 0.1, 1.0\}$) with k-means regime assignments. TEMPO achieves 20–39% lower reconstruction error than the global POD at K=10 modes, with the gap growing in both M and K; no violations of the bound were observed across all $K \in \{1, \ldots, 32\}$.

4.4. Regime Discovery

We select M=3 based on entropy gain: H increases by +0.30 from M=2 to M=3 but only +0.18 from M=3 to M=4, indicating the third regime captures genuine structure that a fourth does not. BIC increases monotonically with M (favoring M=2 by complexity alone), so entropy is the decisive criterion here (Table A.4, Appendix A.3). Figure 3 confirms three distinct manifold branches, one per discovered regime. The discovered regimes capture the joint structure of initial conditions and viscosity in the solution space. The per-viscosity gating weights (Table A.5, Appendix A.3) are nearly uniform across $\nu \in \{0.001, 0.1, 1.0\}$, closely matching the global weights $\boldsymbol{\pi}$=(0.324, 0.267, 0.409): the initial condition determines regime membership, while ν shapes within-regime dynamics; the same initial condition yields a shock at ν=0.001 and a smooth wave at ν=1.0. At inference, ν is passed to the branch network as a conditioning input, capturing both sources of solution variability.

4.5. Role of the Physical Parameter

We ablate the effect of κ at inference by replacing it with zero or a learned prediction. On Burgers', TEMPO(NeuralPOD) barely needs κ: removing it costs only $\sim$1% in L2, since u_0 carries sufficient information to determine regime

Table 1: **Relative L^2 test error on 1D Burgers' equation.** Each per-ν specialist is trained on a single viscosity and evaluated on all three jointly-trained values; underlined entries are in-distribution. The large off-diagonal errors motivate joint training with regime discovery. All jointly-trained methods use $\nu \in \{0.001, 0.1, 1.0\}$. **Bold**: best per column among jointly-trained methods.

Method	Train ν	ν=0.001	ν=0.1	ν=1.0	Infer. (ms)
Per-ν specialists (single-viscosity training)					
DeepONet [3]	0.001	0.536	0.474	0.382	0.037
	0.1	0.535	0.471	0.349	
	1.0	0.547	0.483	0.283	
POD-DeepONet [2]	0.001	0.273	0.297	0.639	0.004
	0.01	0.274	0.274	0.614	
	0.1	0.349	0.048	0.344	
	1.0	0.461	0.260	0.025	
NeuralPOD-DeepONet *(ours)*	0.001	0.392	0.288	0.535	3.53
	0.01	0.391	0.284	0.541	
	0.1	0.426	0.229	0.371	
	1.0	0.483	0.315	0.183	
FNO [9]	0.001	0.426	0.529	0.623	9.90
	0.01	0.454	0.401	0.492	
	0.1	0.650	0.304	0.521	
	1.0	1.161	0.919	0.158	
CNO [10] (2.0M)	0.01	0.687	0.955	2.344	2.14
	0.1	0.983	0.199	0.471	
	1.0	1.870	1.084	0.125	
Joint training ($\nu \in \{0.001, 0.1, 1.0\}$)					
DeepONet [3]	all	0.533	0.472	0.289	0.037
POD-DeepONet [2]	all	0.460	0.339	0.210	**0.004**
FNO [9]	all	0.427	0.298	0.202	9.90
CNO [10] (2.0M)	all	0.361	0.162	**0.133**	2.14
TEMPO(POD) *(ours)*	M=2	0.319	**0.150**	0.159	0.025
	M=3	**0.317**	0.168	0.182	
TEMPO(NeuralPOD) *(ours)*	M=2	0.536	0.458	0.326	0.024
	M=3	0.542	0.456	0.288	

Table 2: **Relative L^2 test error on 2D Darcy flow.** Each per-β specialist is trained on a single source value and evaluated on all five; underlined entries are in-distribution. Off-diagonal errors exceed 1 because the Darcy solution scales with β, so a specialist trained on β=100 predicts amplitudes $\sim 10^4$ times larger than the ground truth at β=0.01. TEMPO trains jointly on $\beta \in \{0.1, 1, 10, 100\}$; the β=0.01 regime is excluded from joint training (—). **Bold**: best per column among jointly-trained methods.

Method	Train β	β=0.01	β=0.1	β=1.0	β=10	β=100	Infer. (ms)
Per-β specialists (single-value training)							
	0.01	4.197	0.615	0.944	0.994	0.999	
	0.1	>10	0.851	0.875	0.987	0.999	
DeepONet [3]	1.0	>10	>1	0.433	0.902	0.972	0.025
	10	>100	>10	>1	0.243	0.705	
	100	>1000	>100	>10	>1	0.078	
	0.01	0.449	0.791	0.961	0.996	>1	
	0.1	>1	0.148	0.871	0.986	0.999	
POD-DeepONet [2]	1.0	>10	>1	0.048	0.897	0.990	0.001
	10	>100	>10	>1	0.035	0.900	
	100	>1000	>100	>10	>1	0.042	
	0.01	1.407	0.786	0.962	0.996	>1	
	0.1	>1	0.421	0.902	0.989	0.999	
NeuralPOD-DeepONet *(ours)*	1.0	>10	>1	0.135	0.901	0.990	0.168
	10	>100	>10	>1	0.050	0.900	
	100	>1000	>100	>10	>1	0.037	
	0.01	0.529	0.791	0.967	0.997	0.9998	
	0.1	6.26	0.144	0.877	0.988	0.999	
FNO [9]	1.0	73.1	8.41	0.034	0.909	0.992	0.587
	10	>100	72.6	8.18	0.013	0.900	
	100	>1000	256.3	43.2	6.13	0.008	
	0.01	0.388	0.929	1.000	1.001	1.000	
CNO [10] (2.0M)	0.1	10.6	0.125	0.948	1.004	1.002	2.24
	1.0	67.0	7.84	0.024	0.896	0.989	
	10	>1000	99.1	8.37	0.009	0.898	
Joint training ($\beta \in \{0.1, 1, 10, 100\}$)							
DeepONet [3]	all	—	76.53	8.503	1.352	0.493	0.025
POD-DeepONet [2]	all	—	2.557	0.363	0.181	0.347	**0.001**
NeuralPOD-DeepONet *(ours)*	all	—	72.9	6.92	0.357	0.345	0.168
FNO [9]	all	—	0.380	0.083	0.023	0.008	0.587
CNO [10] (2.0M)	all	—	**0.133**	**0.026**	**0.008**	**0.005**	2.24
TEMPO(POD) *(ours)*	M=5	—	0.775	0.178	0.136	0.073	0.006
TEMPO(NeuralPOD) *(ours)*	M=5	—	3.813	0.549	0.322	0.311	0.006

Table 3: **Relative L^2 test error on 2D incompressible Navier-Stokes.** Each per-Re specialist is trained on a single Reynolds number and evaluated on all four; underlined entries are in-distribution. Joint methods train on all Re simultaneously. **Bold**: best per column among jointly-trained methods. † kappa-initialised variant.

Method	Train Re	Re=100	Re=1000	Re=3600	Re=10000	Infer. (ms)
Per-Re specialists (single-Re training)						
DeepONet [3]	100	<u>1.000</u>	1.002	1.003	1.004	0.135
	1000	1.000	<u>1.000</u>	1.000	1.000	
	3600	1.002	1.000	<u>1.000</u>	1.000	
	10000	1.001	1.000	1.000	<u>1.000</u>	
POD-DeepONet [2]	100	<u>0.082</u>	0.748	0.865	0.877	0.005
	1000	0.667	<u>0.129</u>	0.671	0.812	
	3600	0.660	0.392	<u>0.180</u>	0.623	
	10000	0.934	0.687	0.556	<u>0.345</u>	
NeuralPOD-DeepONet *(ours)*	1000	0.762	<u>0.198</u>	0.607	0.761	4.87
FNO [9]	100	<u>0.013</u>	0.262	0.368	0.437	0.30
	1000	0.287	<u>0.033</u>	0.138	0.227	
	3600	0.381	0.150	<u>0.039</u>	0.128	
	10000	0.461	0.260	0.126	<u>0.042</u>	
CNO [10]	100	<u>0.014</u>	0.047	0.055	0.058	0.50
	1000	0.027	<u>0.028</u>	0.033	0.035	
	3600	0.030	0.029	<u>0.033</u>	0.035	
	10000	0.030	0.028	0.032	<u>0.034</u>	
Joint training (all Re)						
POD-DeepONet [2]	all	0.092	0.132	0.141	0.145	**0.005**
FNO [9]	all	**0.009**	**0.019**	**0.024**	**0.027**	0.30
NeuralPOD-DeepONet *(ours)*	all	0.700	0.713	0.314	0.320	4.87
CNO [10]	all	0.020	0.035	0.040	0.043	0.50
TEMPO(POD) *(ours)*	M=4	0.097	0.144	0.155	0.170	0.021
TEMPO(NeuralPOD)† *(ours)*	M=4	0.595	0.620	0.694	0.627	0.021

routing on this benchmark. TEMPO(POD) is more sensitive (+37%); however, ν can be predicted from u_0 with 87% accuracy, reducing the gap to +7%. On Darcy, κ is essential for TEMPO(POD): removing β collapses gating to a fixed mixture and increases error by $\sim 3\times$ (+219%); TEMPO(NeuralPOD) is less sensitive (+27%), as its richer nonlinear basis partially compensates. In both cases, β cannot be predicted from the permeability field $a(x)$ because it is drawn independently in the dataset (classifier accuracy 21%, below chance). Full results are in Table A.6 (Appendix A.3).

5. Conclusion

Operator learning for parametric PDEs typically relies on a single global representation. When the governing parameter drives qualitative change in solution structure, a single global basis cannot fit all regimes well and prediction quality suffers.

TEMPO discovers regime structure from solution trajectories via EM and fits a dedicated basis to each regime. The discovered partition captures variability driven by both initial conditions and the physical parameter: on 1D Burgers', regimes persist across all viscosity values, with ν shaping within-regime dynamics; the same initial condition produces a shock at ν=0.001 and a smooth wave at ν=1.0, yet falls in the same regime throughout. A global model cannot separate these sources of variation; TEMPO recovers their structure from trajectory data without regime-label supervision.

TEMPO(POD) outperforms the joint POD-DeepONet baseline by 13–50% on Burgers' and up to $5\times$ on Darcy flow, where per-parameter specialists fail catastrophically out-of-distribution. On Navier-Stokes, solutions at different Re share similar POD directions, and TEMPO correctly recovers the global baseline without imposing spurious structure. TEMPO(NeuralPOD) extends the framework to nonlinear bases; improving nonlinear basis fitting within the EM loop remains an open direction.

Several directions remain open. Conditioning basis functions on κ would let each mode encode parameter-dependent dynamics directly. Extending regime discovery to continuous parameter spaces and three-dimensional problems is a natural next step.

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Appendix A.

Appendix A.1. Proof of Proposition 1

Lemma 1. *Let $\bar{s} = \frac{1}{N}\sum_i s^{(i)}$ be the global mean and $\bar{s}_m = \frac{1}{N_m}\sum_i \gamma_{im} s^{(i)}$ the regime-m mean, with $N_m = \sum_i \gamma_{im}$. For any rank-K orthoprojector P and any regime m,*

$$\sum_i \gamma_{im} \|(I-P)(s^{(i)}-\bar{s}_m)\|_w^2 = \sum_i \gamma_{im} \|(I-P)(s^{(i)}-\bar{s})\|_w^2 - N_m \|(I-P)(\bar{s}_m-\bar{s})\|_w^2.$$

Proof. Let $c_m = \bar{s}_m - \bar{s}$. Writing $(I-P)(s^{(i)}-\bar{s}_m) = (I-P)(s^{(i)}-\bar{s}) - (I-P)c_m$ and expanding $\|a-b\|_w^2 = \|a\|_w^2 - 2\langle a,b\rangle_w + \|b\|_w^2$, then summing over i with weights γ_{im}:

$$\begin{aligned}\sum_i \gamma_{im} \|(I-P)(s^{(i)}-\bar{s}_m)\|_w^2 = &\sum_i \gamma_{im} \|(I-P)(s^{(i)}-\bar{s})\|_w^2 \\ &- 2\Big\langle (I-P)\sum_i \gamma_{im}(s^{(i)}-\bar{s}),\ (I-P)c_m \Big\rangle_w \\ &+ N_m \|(I-P)c_m\|_w^2.\end{aligned}$$

Since $\sum_i \gamma_{im}(s^{(i)}-\bar{s}) = N_m c_m$, the cross-term equals $-2N_m\|(I-P)c_m\|_w^2$; together with the quadratic term this gives $-N_m\|(I-P)c_m\|_w^2 = -N_m\|(I-P)(\bar{s}_m-\bar{s})\|_w^2$. □

Proof of Proposition 1. Let P^* and P_m^* be the globally and regime-m optimal rank-K projectors. Since P_m^* minimizes the regime-m objective over all rank-K projectors, substituting the (generally suboptimal) P^* cannot decrease it:

$$\varepsilon_{\text{TEMPO}}^K = \sum_m \sum_i \gamma_{im} \|(I-P_m^*)(s^{(i)}-\bar{s}_m)\|_w^2 \ \le\ \sum_m \sum_i \gamma_{im} \|(I-P^*)(s^{(i)}-\bar{s}_m)\|_w^2.$$

Applying Lemma 1 to each m and summing over m, then using $\sum_m \gamma_{im} = 1$ for each i:

$$\begin{aligned}\sum_m \sum_i \gamma_{im} \|(I-P^*)(s^{(i)}-\bar{s}_m)\|_w^2 &= \sum_i \|(I-P^*)(s^{(i)}-\bar{s})\|_w^2 - \underbrace{\sum_m N_m \|(I-P^*)(\bar{s}_m-\bar{s})\|_w^2}_{\ge 0} \\ &\le \sum_i \|(I-P^*)(s^{(i)}-\bar{s})\|_w^2 = \varepsilon_{\text{global}}^K.\end{aligned}$$

□

Appendix A.2. 2D Navier-Stokes Data Generation

We use the same equation and forcing as in [9] with a pseudo-spectral solver in stream-function formulation: diffusion treated implicitly (Crank–Nicolson), advection explicitly (second-order Adams–Bashforth), and $\frac{2}{3}$-rule dealiasing on the 64×64 grid. Simulation time-step δt=0.01; each recorded snapshot aggregates 10 sub-steps (Δt=0.1). Initial vorticity fields are random sums of sinusoids,

$$\omega_0(\mathbf{x}) = \sum_{k=1}^{7} \frac{a_k}{1+k}\left[\sin(2\pi k\, x_1+\phi_k)+\cos(2\pi k\, x_2+\phi_k)\right], \quad a_k \sim \mathcal{N}(0,1),\ \phi_k \sim \mathcal{U}[0,2\pi].$$

A 5 s burn-in (500 simulation steps) discards transient behavior before recording begins.

Appendix A.3. Additional Experiments

Regime model selection (1D Burgers'). Table A.4 reports BIC, mean assignment entropy H, and mean test error $\bar{\varepsilon}$ for $M \in \{2,3,4\}$. Table A.5 shows the per-viscosity EM responsibilities alongside global mixture weights π_m.

Table A.4: Regime count selection on 1D Burgers'. BIC $= -2\log\mathcal{L} + k\log N$ (lower is better; penalizes complexity); H: mean assignment entropy (higher gain signals a new genuine regime); $\bar{\varepsilon}$: mean relative L^2 test error. BIC increases monotonically, favoring M=2 by complexity alone. **Bold row**: selected configuration (M=3), chosen because the entropy gain ΔH=0.30 from M=2→3 substantially exceeds ΔH=0.18 from M=3→4.

M	BIC ($\times 10^4$, ↓)	H	$\bar{\varepsilon}$
2	3.90	0.622	0.209
3	**4.92**	**0.920**	0.222
4	5.47	1.099	0.228

Table A.5: Mean EM assignment weights per viscosity (1D Burgers', TEMPO(POD), M=3) vs. global mixture weights π_m. Each row sums to 1.

ν	Regime 1	Regime 2	Regime 3
0.001	0.32	0.28	0.40
0.1	0.32	0.26	0.42
1.0	0.33	0.26	0.41
π_m	0.32	0.27	0.41

Role of the physical parameter (ablation). Table A.6 reports the effect of replacing κ with zero or a learned prediction at inference.

POD vs. NeuralPOD basis comparison (1D Burgers'). Figure A.4 shows the singular-value and residual decay for POD (K=30 modes capture 99.99% of the variance) and Fourier NeuralPOD. The corresponding spatiotemporal mode functions (Figure A.5) reveal a structural difference: POD modes are globally supported, while NeuralPOD modes localise adaptively — at ν=0.01 they concentrate near the shock front. Per-sample reconstructions from both bases are compared in Figure A.7; additional NeuralPOD examples are shown in Figure A.6.

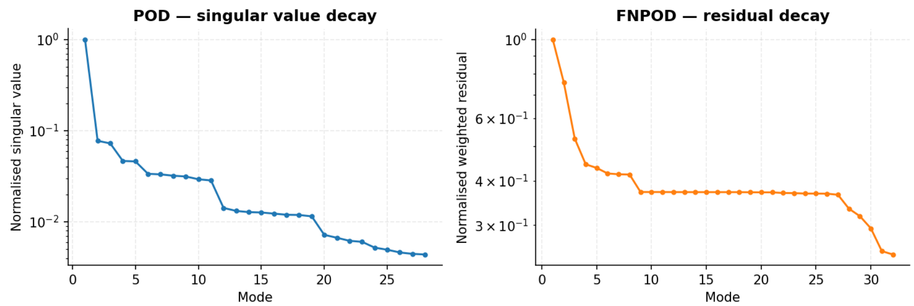
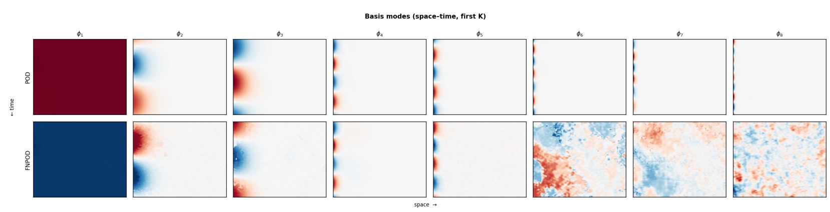
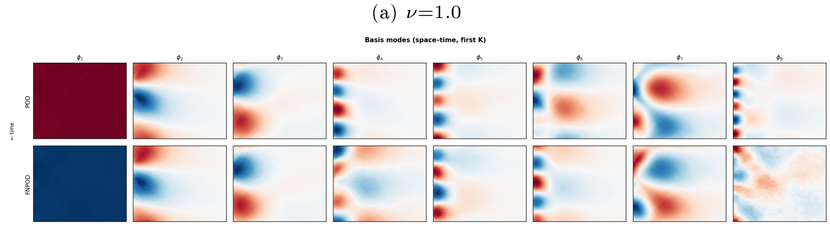


Figure A.4: Basis convergence on 1D Burgers' (ν=1.0): POD singular-value decay (*left*); Fourier NeuralPOD residual decay (*right*).



(a) ν=1.0



(b) ν=0.01

Figure A.5: First K basis modes on 1D Burgers': POD (*top row*) vs. Fourier NeuralPOD (*bottom row*).

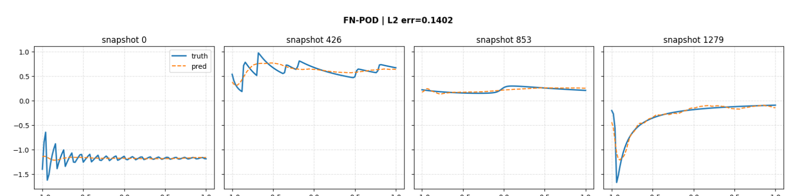


Figure A.6: Fourier NeuralPOD representative reconstructions on 1D Burgers' (ν=1.0).

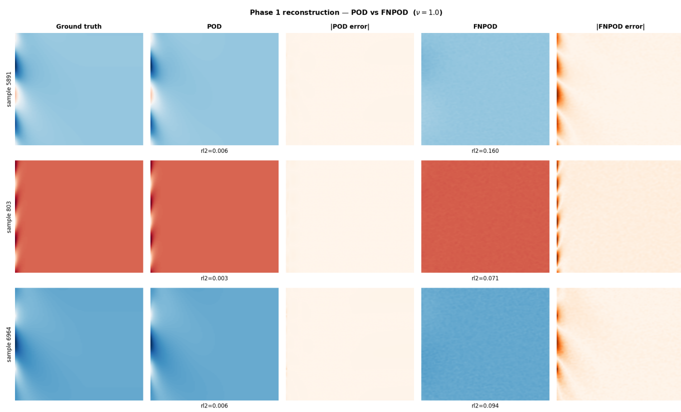
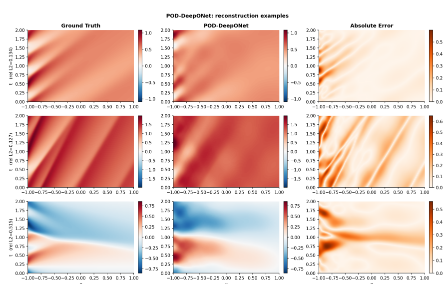
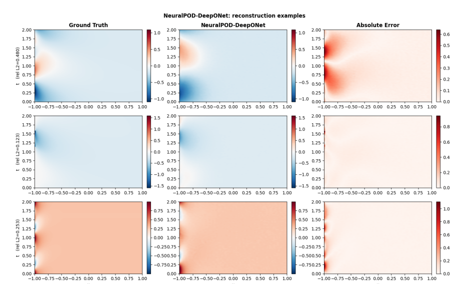


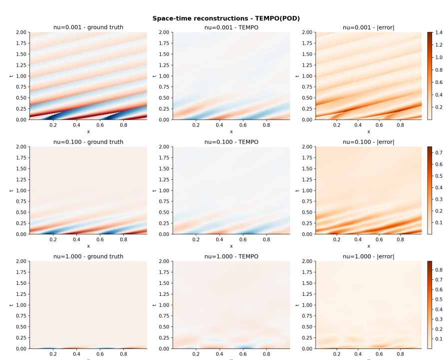
Figure A.7: Phase 1 reconstruction comparison on 1D Burgers' (ν=1.0). Each row shows one sample: ground truth, POD reconstruction, |POD error|, Fourier NeuralPOD reconstruction, |FNPOD error|.



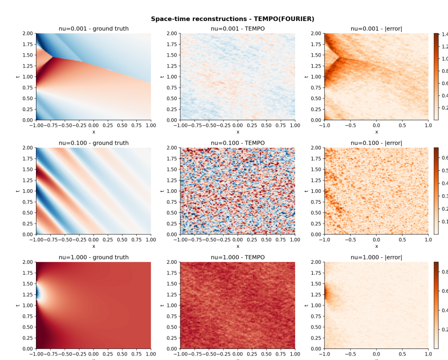
(a) POD-DeepONet



(b) NeuralPOD-DeepONet



(c) TEMPO(POD), M=3



(d) TEMPO(NeuralPOD), M=3: high-frequency noise (L2 0.46–0.54)

Figure A.8: Space–time reconstructions on 1D Burgers' (ν=1.0).

Table A.6: κ ablation: increase in mean relative L2 error when κ is removed or predicted, relative to the real-κ baseline. For Darcy, β is drawn independently of $a(x)$; prediction from input data is not possible.

	Burgers'		Darcy	
	NPOD (M=3)	POD (M=3)	NPOD (M=5)	POD (M=5)
$\kappa = 0$	+0.006 (+1%)	+0.127 (+37%)	+0.334 (+27%)	+0.618 (+219%)
κ predicted from u_0	—	+0.024 (+7%)	not predictable	

Representative reconstructions (1D Burgers'). Figure A.8 shows space–time predictions for all four methods on 1D Burgers' (ν=1.0). Each panel: ground truth (*left*), prediction (*center*), point-wise error (*right*).

TEMPO regime structure across benchmarks. Figure A.10 compares the Phase 1 regime structure across all three benchmarks. On Burgers' (NeuralPOD), the partition mirrors the POD variant (Figure 3) with sharper assignments. On Darcy flow, regimes span multiple β values; TEMPO(NeuralPOD) finds a nearly identical partition. On Navier-Stokes, all Re values overlap in POD coefficient space and regimes 2–4 collapse to a few points, explaining why dedicated bases bring no gain.

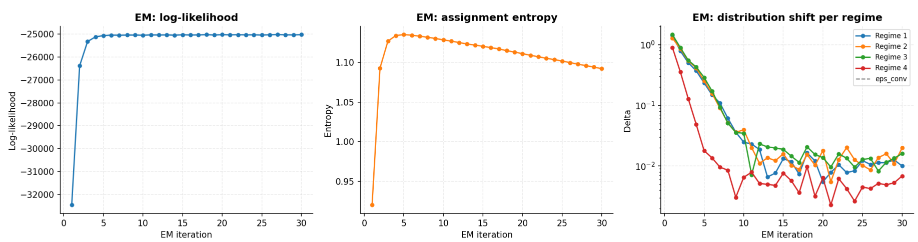
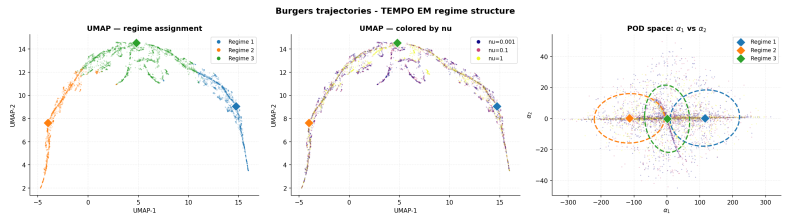


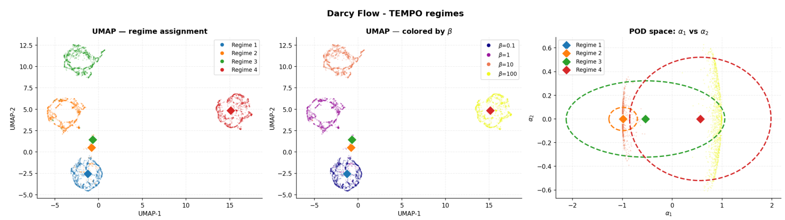
Figure A.9: TEMPO(POD) EM convergence on 2D Navier-Stokes (M=4).

TEMPO reconstructions (2D Darcy flow). Figure A.11 shows representative predictions from both TEMPO variants on 2D Darcy flow (M=5). Each panel: ground truth (*left*), prediction (*center*), point-wise error (*right*).

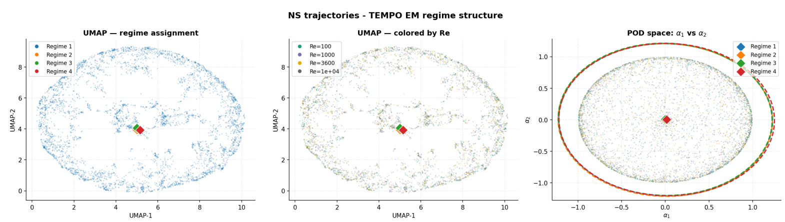
TEMPO reconstructions (2D Navier-Stokes). Figure A.12 shows representative predictions from TEMPO(POD) and TEMPO(NeuralPOD) on 2D Navier-Stokes (M=4); EM convergence is shown in Figure A.9.



(a) 1D Burgers', NeuralPOD (M=3, H=0.23)

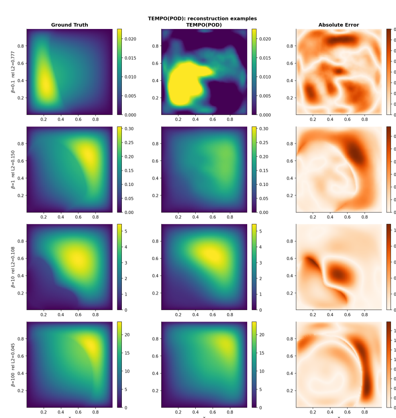


(b) 2D Darcy flow, POD (M=4)

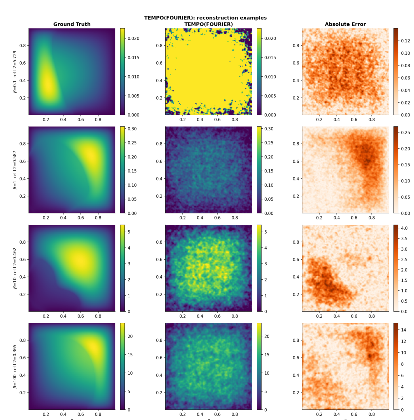


(c) 2D Navier-Stokes, POD (M=4): no Re separation

Figure A.10: TEMPO EM regime structure across all benchmarks. Layout as in Figure 3.

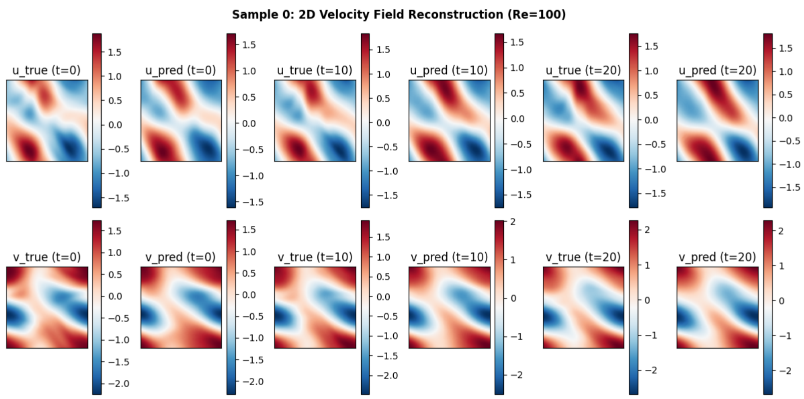


(a) TEMPO(POD)

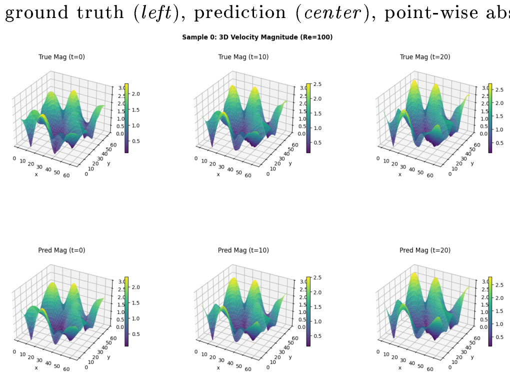


(b) TEMPO(NeuralPOD)

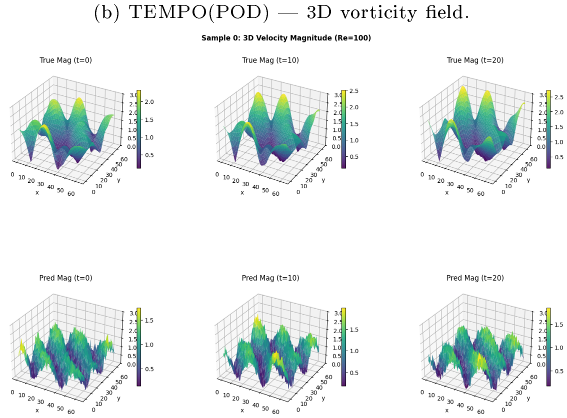
Figure A.11: Space–time reconstructions on 2D Darcy flow (M=5).



(a) TEMPO(POD): ground truth (*left*), prediction (*center*), point-wise absolute error (*right*).



(b) TEMPO(POD) — 3D vorticity field.



(c) TEMPO(NeuralPOD) — 3D vorticity field.

Figure A.12: 2D Navier-Stokes (M=4): TEMPO(POD) prediction (*top*); TEMPO(POD) and TEMPO(NeuralPOD) 3D vorticity (*middle*, *bottom*).

Algorithm 1 NeuralBasis

Require: weighted snapshots $\{s^{(i)}, \gamma_{im}\}_{i=1}^{N}$, tolerance τ

Ensure: basis $\phi_0^{(m)}$, $\{\phi_k^{(m)}, \{\lambda_k^{(m,i)}\}_{i=1}^{N}\}_{k=1}^{K_m}$

$\bar{s}^{(m)} \leftarrow \dfrac{1}{N_m} \sum_{i=1}^{N} \gamma_{im}\, s^{(i)}, \quad N_m = \sum_{i=1}^{N} \gamma_{im}$ ▷ responsibility-weighted mean trajectory

Train $\phi_0^{(m)}$ by minimizing $\sum_j w_j \big(\phi_0^{(m)}(y_j) - \bar{s}_j^{(m)}\big)^2$

$r^{(0,i)} \leftarrow s^{(i)} - \phi_0^{(m)}(y)$ for all i; $k \leftarrow 0$

$V_m \leftarrow \sum_{i=1}^{N} \gamma_{im} \, \|s^{(i)} - \bar{s}^{(m)}\|_w^2$ ▷ initial unmodeled variance

while $\sum_{i=1}^{N} \gamma_{im} \, \|r^{(k,i)}\|_w^2 \ \geq \ \tau V_m$ **and** $k < K_{\max}$ **do**

  Jointly minimize over $\phi_{k+1}^{(m)}$ and $\{\lambda_{k+1}^{(m,i)}\}_{i=1}^{N}$:

$$\sum_{i=1}^{N} \gamma_{im} \sum_j w_j \Big(\lambda_{k+1}^{(m,i)}\, \phi_{k+1}^{(m)}(y_j) - r_j^{(k,i)}\Big)^2$$

  $r^{(k+1,i)} \leftarrow r^{(k,i)} - \lambda_{k+1}^{(m,i)}\, \phi_{k+1}^{(m)}(y)$ for all i

  $k \leftarrow k+1$

end while

return $K_m \leftarrow k$, $\phi_0^{(m)}$, $\{\phi_k^{(m)}, \{\lambda_k^{(m,i)}\}_{i=1}^{N}\}_{k=1}^{K_m}$

Algorithm 2 TEMPO

Require: $\mathcal{D} = \{u_0^{(i)}, \kappa^{(i)}, s^{(i)}\}_{i=1}^{N}$; M, ε_s, ε_l, ε_p, ε_c, τ

Ensure: Regime bases $\{\Phi_m^*\}$, amplitudes $\{\Lambda_m^*\}$, responsibilities $\{\gamma_{im}^*\}$, weights $\{\pi_m^*\}$

Initialization

Compute global POD on $\{s^{(i)}\}$; project to $\alpha^{(i)} \in \mathbb{R}^{d_0}$ for all i

Run GMM-EM on $\{\alpha^{(i)}\} \to \gamma_{im}^{(0)}$, $\pi_m^{(0)}$, $\mu_m^{(0)}$, $\Sigma_m^{(0)}$

for $m = 1, \ldots, M$ **do**

  $\Phi_m^{(0)}, \Lambda_m^{(0)} \leftarrow$ NeuralBasis($\{s^{(i)}, \gamma_{im}^{(0)}\}, \tau$) ▷ or weighted SVD for TEMPO(POD), see §3.1

end for

$\sigma^2 \leftarrow (2N)^{-1}\sum_{i,m} \gamma_{im}^{(0)} \, \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2$ (14)

EM iterations

repeat

  E-step

  for all $i = 1, \ldots, N$ and $m = 1, \ldots, M$ **do**

    Compute $\hat{s}_m^{(i)}$ via (4) using $\Phi_m^{(q-1)}$ and $\Lambda_m^{(q-1)}$

    $\gamma_{im}^{(q)} \leftarrow$ Eq. (15)

  end for

  M-step: mixture weights

  $\pi_m^{(q)} \leftarrow N_m^{(q)}/N$ for all m (16)

  M-step: bases

  for $m = 1, \ldots, M$ **do**

    Compute $\mu_m^{(q)}$ (18), $\Sigma_m^{(q)}$ (19), $\Delta_m^{(q)}$ (20)

    if $\Delta_m^{(q)} < \varepsilon_s$ **then**

      $\Phi_m^{(q)} \leftarrow \Phi_m^{(q-1)}$ Skip

    else if $\Delta_m^{(q)} < \varepsilon_l$ **then** Incremental

      Compute $r_m^{(K_m)}$ under updated $\gamma^{(q)}$; $V_m^{(q)} \leftarrow \sum_i \gamma_{im}^{(q)} \|s^{(i)} - \bar{s}_m^{(q)}\|_w^2$

      if $\sum_i \gamma_{im}^{(q)} \|r_m^{(K_m,i)}\|_w^2 \geq \tau \cdot V_m^{(q)}$ **then**

        Train mode $K_m{+}1$; $K_m \leftarrow K_m + 1$

      end if

      if $\|\phi_{K_m}^{(m)}\|_w^2 \cdot \sum_{i=1}^{N} \gamma_{im}^{(q)} \left(\lambda_{K_m}^{(m,i)}\right)^2 < \varepsilon_p$ **then**

        $K_m \leftarrow K_m - 1$

      end if

    else Full rerun

      $\Phi_m^{(q)}, \Lambda_m^{(q)} \leftarrow$ NeuralBasis($\{s^{(i)}, \gamma_{im}^{(q)}\}, \tau$)

    end if

  end for

until $\max_m \Delta_m^{(q)} < \varepsilon_c$

return $\Phi_m^* \leftarrow \Phi_m^{(q)}$, $\Lambda_m^* \leftarrow \Lambda_m^{(q)}$, $\gamma_{im}^* \leftarrow \gamma_{im}^{(q)}$, $\pi_m^* \leftarrow \pi_m^{(q)}$